

Jarvis Voice Assistant - Setup Guide

Raspberry Pi 5 + Hermes Agent + OpenWakeWord + Whisper

1. System Dependencies

```
bash
sudo apt install portaudio19-dev -y
```

2. Python Packages

```
bash
pip install openwakeword faster-whisper pyaudio --break-system-packages
```

3. Verify Audio Device

```
bash
# List capture devices – note the card number for your mic
arecord -l

# List PyAudio device indices
python -c "import pyaudio; p=pyaudio.PyAudio(); [print(i, p.get_device_info_by_index(i).name) for i in range(p.get_device_count())]"

# Check supported sample rate
arecord -D hw:<CARD>,0 --dump-hw-params /dev/null 2>&1 | grep -i rate
```

4. Verify Wake Word Model

```
bash
# Should show hey_jarvis_v0.1.onnx in the list
ls ~/.local/lib/python3.13/site-packages/openwakeword/resources/models/
```

5. Place the Script

```
bash
```

```
# Copy jarvis.py to home directory
cp jarvis.py ~/jarvis.py
chmod +x ~/jarvis.py
```

6. Update jarvis.py Config if Needed

Key variables at the top of the script:

- `DEVICE_INDEX` — PyAudio index of your mic (check step 3)
- `DEVICE_RATE` — sample rate of your device (48000 for USB-C Speaker)
- `WAKEWORD_MODEL` — path to the .onnx model (check step 4)
- `WAKEWORD_THRESHOLD` — sensitivity 0.0-1.0 (default 0.5)
- `WHISPER_MODEL` — tiny/base/small (tiny is fastest on Pi)

7. Run

```
bash

python ~/jarvis.py
```

Notes

- The ALSA warnings on startup are harmless — ignore them
- Wake word: **"Hey Jarvis"**
- Hermes must be installed and configured (`hermes_model` set to LM Studio endpoint)
- LM Studio must be running on your GPU machine with the server enabled
- To change wake word, swap the .onnx model path — available models:
 - hey_jarvis, hey_marvin, hey_mycroft, alexa, weather, timer

Auto-start on Boot (optional)

```
bash
```

```
# Create a systemd user service
mkdir -p ~/.config/systemd/user
cat > ~/.config/systemd/user/jarvis.service << 'EOF'
[Unit]
Description=Jarvis Voice Assistant
After=network.target

[Service]
ExecStart=/usr/bin/python3 /home/%u/jarvis.py
Restart=on-failure
RestartSec=5

[Install]
WantedBy=default.target
EOF

systemctl --user enable jarvis.service
systemctl --user start jarvis.service
```